

Revised Generation Interconnection System Impact Study Report

For Queue Project Q14-0321

Golden Ring 138 kV
November 19, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0321. The Interconnection Customer proposes a 159.4 MW battery energy storage facility interconnecting at the Golden Ring 138 kV substation in the Virginia area. The request was received on March 23, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 174 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,438,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Myrtle / Sanger 138kV, the Segs 6 / Golden Ring 138kV loads to 161.6% of its 217 MVA rating. Project DFAX: 100.0%.

For the loss of Myrtle / Sanger 138kV Line, the Werner Est / Segs 6 138kV loads to 161.3% of its 221 MVA rating. Project DFAX: 100.0%.

The Myrtle / Sanger 138kV line is overloaded to 154.8% of its 230 MVA emergency rating for the loss of the Segs 6 / Werner W 138kV L. The project contributes 100.0% DFAX (159.4 MW) to this overload.

The Suffok / Myrtle 138kV line is overloaded to 150.3% of its 213 MVA emergency rating for the loss of the Werner W / Segs 6 138kV L. The project contributes 100.0% DFAX (159.4 MW) to this overload.

Suffolk / Ltg Mcconal 138kV: 144.9% loading (200 MVA) following the outage of the Segs 6 / Werner W 138kV; DFAX = 100.0%.

Ltg Mcconal / Golden Ring 138kV: 129.8% loading (201 MVA) following the outage of the Segs 6 / Werner W 138kV; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Segs 6 / Golden Ring Substation 138kV L	Rebuild line with higher-capacity conductor	\$11,361,000	72% (\$8,164,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Werner W / Segs 6 138kV L	Rebuild line with higher-capacity conductor	\$12,221,000	72% (\$8,783,000)
Myrtle / Sangr 138kV	Rebuild line with higher-capacity conductor	\$12,846,000	72% (\$9,232,000)
Myrtle / Suffolk 138kV	Rebuild line with higher-capacity conductor	\$4,237,000	72% (\$3,045,000)
Suffolk / Ltg Mcconal 138kV	Rebuild line with higher-capacity conductor	\$4,288,000	72% (\$3,081,000)
Ltg Mcconal / Golden Ring 138kV	Rebuild line with higher-capacity conductor	\$2,968,000	72% (\$2,133,000)
	Total allocated to project		\$34,438,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.